

Roll No.

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BT/AI/CS/W-25/AI/CS5M003



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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU



~~B. Tech Summer / Winter – 2025-26~~
(ESE / Make-up / Supplementary) Examination

Branch: Artificial Intelligence / CS

Sem: -5th

Subject: LLM & Prompt Engineering

Subject Code: AI5M003 / CS5M003

Marks: 100

Date of Exam: 14/11/25

Time: 3 Hrs.

Instructions to the students:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

1. Understand and apply basic and advanced prompt engineering techniques.
2. Integrate and utilize APIs to build AI applications.
3. Design and implement AI-powered Chatbot's and utilities.
4. Optimize generative AI applications for latency and cost.
5. Evaluate prompt performance and select appropriate fallback strategies.

SECTION A

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Prompt Engineering.	L1/CO1	2
Q.2	State what is meant by Generative AI.	L1/CO1	2
Q.3	Illustrate the purpose of a text summarizer in AI systems.	L3/CO4	2
Q.4	List out two sampling methods used in prompt engineering.	L1/CO2	2
Q.5	Write a short note on Chain-of-Thought prompting in language models.	L2/CO2	2
Q.6	Classify an API key in OpenAI API authentication.	L2/CO3	2
Q.7	Restate the primary purpose of the Hugging face Model.	L2/CO3	2
Q.8	Write the purpose of the Chat endpoint in OpenAI.	L3/CO4	2
Q.9	Give an example of a UI framework used for AI applications.	L2/CO4	2
Q.10	Define multi-model orchestration.	L1/CO5	2
Q.11	State the function of cost estimation in token-based billing.	L1/CO5	2
Q.12	Enlist components of Lang chain model.	L1/CO3	2

SECTION B

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Distinguish between Traditional AI and Generative AI.	L4/CO1	4
Q.14	Illustrate the example of Persona creation and show how they affect model responses.	L3/CO2	4
Q.15	Discuss any 3 components in the Hugging Face model for a large language model.	L2/CO3	4
Q.16	Explain the difference between stateless and stateful chatbots.	L5/CO4	4
Q.17	Write a short note on performance profiling and its significance.	L3/CO5	4
Q.18	Sketch a simple diagram illustrating latency measurement in an AI system.	L3/CO5	4

SECTION C

1. Solve any FOUR Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate the example & concept of Chain-of-Thought prompting and also show where it enhances complex problem-solving.	L4/CO2	7
Q.20	Explain in detail how to design and build an AI-based text summarizer using NLP techniques.	L5/CO4	7
Q.21	Distinguish between FAISS and Chroma DB based on their architecture, purpose, and usage.	L4/CO3	7
Q.22	Examine the working of AI-based translators and the challenges involved in language translation.	L4/CO4	7
Q.23	Describe prompt tuning and how it leverages embeddings to customize model behaviour without full fine-tuning.	L5/CO2	7
Q.24	Discuss prompt relevance, coherence, and accuracy with proper illustrations and use-case examples.	L5/CO5	7

SECTION D

1. Solve any TWO Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Explain about following in brief: a) Re-Act & Self-Ask. b) Hugging Face. c) FAISS & JSON. d) Chat endpoints.	L5/CO3	12
Q.26	Justify the architecture of a Transformer-based Large Language Model (LLM) in brief.	L6/CO1	12
Q.27	Design an effective prompt design pattern incorporating techniques such as Re-Act and Self-Ask. Explain how these patterns enhance the reasoning capabilities and interpretability of large language models	L6/CO2	12